

06

Wear Failure

Comprehensive Study Notes

Course Instructor

Dr. Hanadi Salem

[linkedin.com/in/hanadi-salem-11361a11](https://www.linkedin.com/in/hanadi-salem-11361a11)

Compiled by

Abdelrahman Sawy

[linkedin.com/in/abdelrahman-sawy-542740295](https://www.linkedin.com/in/abdelrahman-sawy-542740295)

The American University in Cairo (AUC) — School of Sciences and Engineering

WEAR FAILURE

Wear is defined as damage to a solid surface caused by the removal or displacement of material by the mechanical action of a contacting solid, liquid, or gas. It can also be defined as “loss of dimension from plastic deformation if it is originated at the interface between two sliding surfaces. However, plastic deformation such as yield stress is excluded from the wear definition if it doesn't incorporate a relative sliding motion and contact against another surface despite the possibility for material removal, because it then lacks the relative sliding action of another surface. Impact wear is a short sliding motion in which “two solid bodies interact at an exceptional short time interval”.

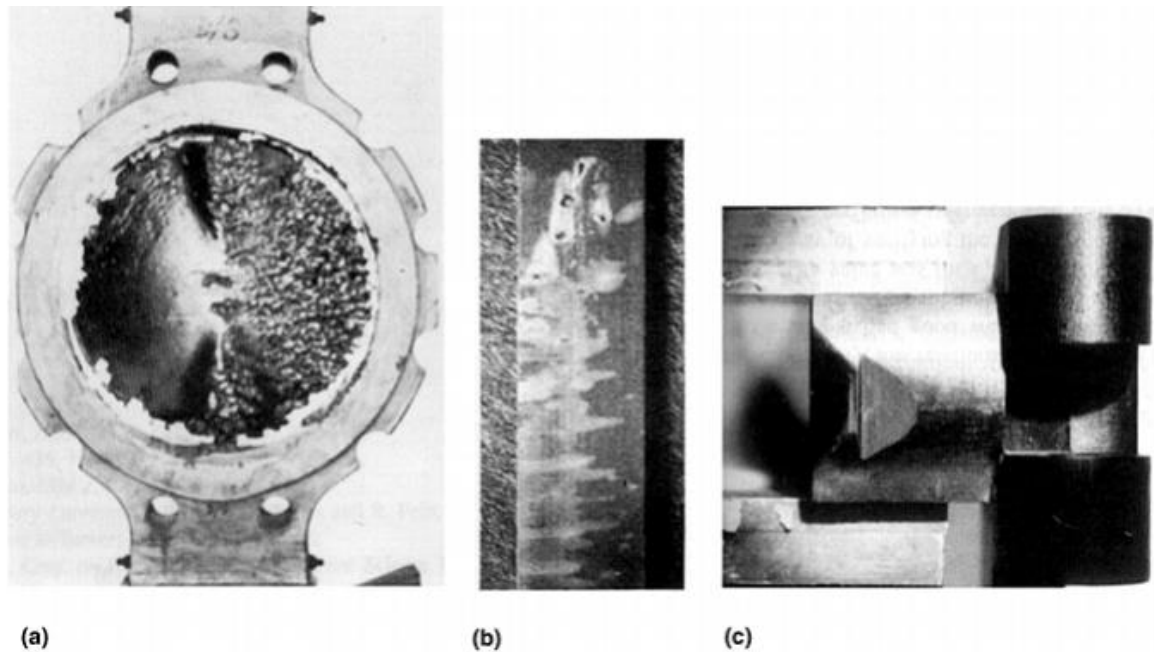


Fig. 1 Examples of wear showing loss of material, changes in dimension, and changes in appearance. (a) Erosion damage on a butterfly valve component. (b) Fretting damage on a friction band. (c) Sliding wear on a cam follower

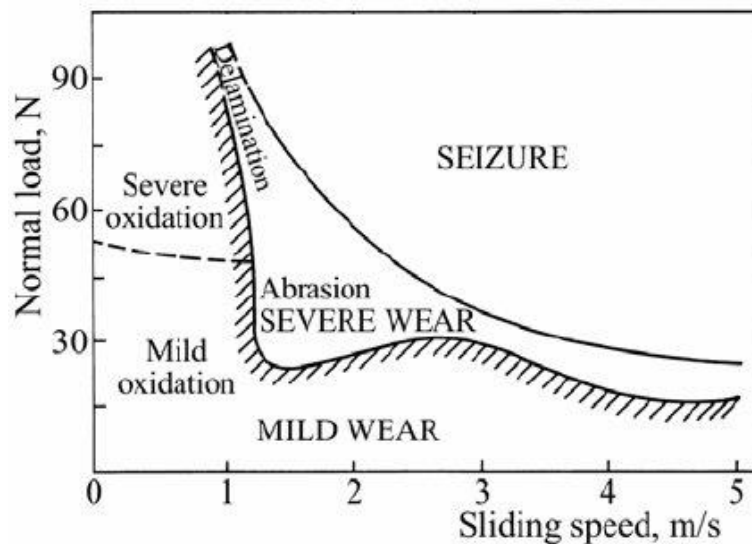
Wear Stages

Under normal mechanical and practical procedures, the wear-rate normally changes through three different stages

- **Primary stage** or early run-in period, where surfaces adapt to each other, and the wear-rate might vary between high and low.
- **Secondary stage** or mid-age process, where a steady rate of ageing is in motion. Most of the components' operational life is comprised in this stage. The secondary stage is shortened with increasing severity of environmental conditions such as higher temperatures, strain rates, stress and sliding velocities etc.
- **Tertiary stage** or old-age period, where the components are subjected to rapid failure due to a high rate of ageing.

Wear Rate Influencing Parameters

- The wear rate is strongly influenced by the operating conditions. Specifically, normal loads and sliding speeds play a pivotal role in determining wear rate.
- Tribo-chemical reaction is also important in order to understand the wear behavior.
- Different oxide layers are developed during the sliding motion.
- Other layers are originated from complex interaction among surface, lubricants, and environmental molecules.
- In general, a single plot, namely wear map demonstrating wear rate under different loading condition is used for operation.
- Higher pressure causes higher wear
- Higher speed causes higher wear
- Heat generated causes material to be softer that causes higher wear
- Higher hardness decreases the wear. Usually, the less hard material would be the one to erode (picked by design)
- Harsh Environmental conditions causes higher wear



The empirical wear-mechanism map for Al alloy 6061 reinforced with SiC whiskers Figure shows how the normalized wear rate and normalized coefficient of friction depend on volume fraction of reinforcement in different Al alloys. Normalized wear rate (composite wear rate / matrix wear rate) decreases with the increase of reinforcement volume fraction, and limit after which this becomes apparent. Similar dependence on reinforcement volume fraction is obtained for the normalized coefficient of friction. Increase of reinforcement size reduces the wear rate as well.

Book Introduction:

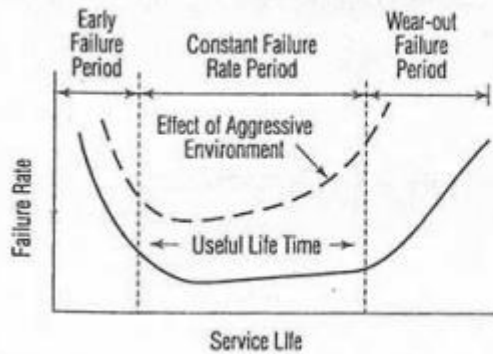


Fig. 1.1 Schematic bath tub curve depicting failure rate as function of service life

Wear is a surface phenomenon occurring as a damage to a solid surface. It is caused by the removal or displacement of material from the surface by the mechanical action of a contacting solid/lubricant, liquid, or gas. Unlike fatigue and corrosion, wear is less catastrophic but greatly reduces the operating efficiency. Wear failure depends on the degree of the wear which in turn is governed by the change in load, speed, or environmental condition. The types of wear generally encountered in engineering systems fall into five distinct categories (a) abrasive; (b) adhesive; (c) erosive; (d) corrosive; (e) erosion-corrosion. Of these abrasive and adhesive modes of failures are common in the aircraft industry.

Abrasive Wear:

It is a form of displacement of the metal from the surface due to contact with high spots or asperities on the mating surface or due to loose hard particles. In the case of hard particles, they may either be trapped in between the two sliding surfaces resulting in abrasion of one or both of them and embedded in either of the surfaces abrading the opposing surface. In abrasive wear also, material is removed from the surface by the cutting action of abrasive particles. Characteristics:

- Wear may range from the mild polishing type to the rapid and severe removal of material with accompanying surface roughening.
- Fairly regular grooves in the smooth surface with or without loosely attached metallic debris, producing damages like.
- Scoring or gouging abrasion identified by the removal of large particles with heavy gouge on the worn surface.
- High stress grinding abrasion by progressive fragmentation of abrasion particles.
- Low stress scratching abrasion caused by scratching of the metal surface.

Adhesive Wear:

It occurs when two metallic surfaces slide against each other under high localized pressure and due to which microscopic projection/ asperities get bonded at the sliding interface. The sliding forces simultaneously break up the bond at the weakest point, resulting in a metal tear from the surface and transference of the same to the other surface. Adhesive wear is common under dry contact or non-lubricated conditions. Service defects commonly encountered due to this type of wear are surface fatigue, spalling, seizing, scuffing, scoring. Characteristics:

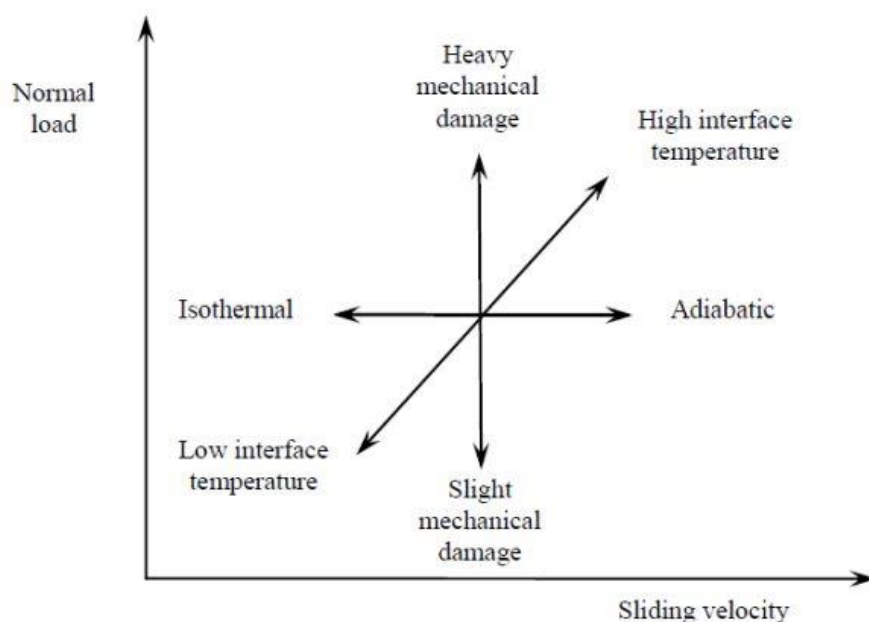
- Exhibits a torn/ dragged and furrowed appearance with marking in the direction of sliding.
- Signs of severe scuffing in the zone of sliding contact.

Types of Wear Process

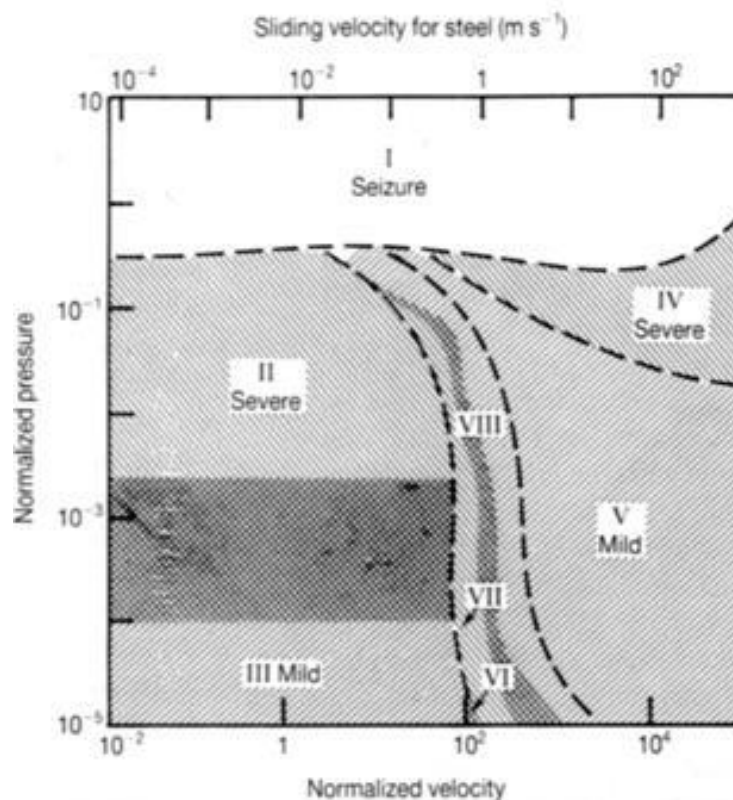
Wear is the deformation and removal of material from its original position on a surface as a result of mechanical action of another surface and/or particles. In general material is removed from a softer surface by a harder surface. There are two main categories for types of wear process, sliding wear and wear by hard particles. The distinction between the two is not sharp, and there will almost always be a degree of both occurring.

1. Sliding Wear

- Occurs when two solid bodies slide over each other. One or both of the surfaces will suffer wear. An example is tires in contact with road.
- Lubrication can dramatically reduce wear rates. Strong interfacial bonds form across asperity junctions. When two dissimilar metals slide against each other the asperity junctions formed are stronger than the weaker of the two metals. This leads to the plucking out of fragments of the softer metal, giving rise to severe wear of the softer metal.
- Extent of wear:
 - Increasing the load leads directly to higher stress, which results in greater wear.
 - Sliding velocity determines the relative rate of heat conduction away from the surface.
 - At low sliding velocity, the heat generated (due to friction) will be relatively rapidly conducted away so the interface temperature stays low (isothermal). At high velocity, only limited heat conduction can occur, so interface temperature increases and the conditions (adiabatic).
 - High interface temperatures increase reactivity of the surfaces, causing rapid growth of oxide films. It also reduces the mechanical strength of asperities and may even cause melting in extreme cases.
 - When the sliding speed is low there is time for heat generated to be conducted to the environment, so the retained heat is low and almost constant (isothermal)
 - If the speed is high heat keeps on increasing (adiabatic) causing higher wear rate.

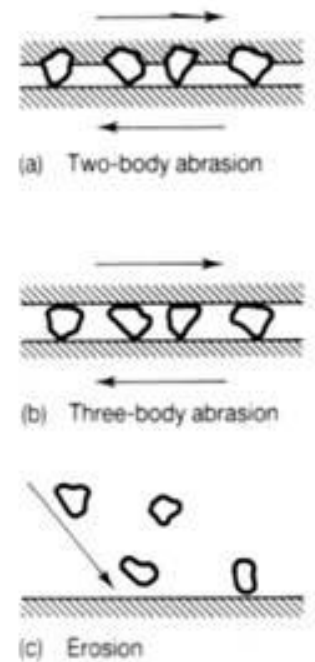


- Mechanism (Wear regime map for the sliding of steel on steel):
 - Regime I: Very high contact pressure. Gross seizure of the surfaces: catastrophic growth of the asperity junctions occurs, leading to the real area of contact becoming equal to the apparent area.
 - Regime II: High loads and relatively low sliding velocity. Penetration of the thin native surface oxide film occurs, leading to high wear rates and metallic debris. Thermal effects are negligible as the sliding velocity is low.
 - Regime III: Lower loads than regime II, resulting in the oxide not being penetrated. Wear is mild because only oxide debris is formed.
 - Regime IV: High loads and sliding speeds. Melting occurs as frictional power dissipation is high and thermal conduction is ineffective at removing heat from the interface. The wear rate is high, with metal being removed as metallic droplets.
 - Regime V: Low contact pressure but high sliding speed. The interface temperature is still high but below the melting point so surface oxidation occurs rapidly. Wear is mild because the debris is oxide.
 - Regime VI: Hot-spots at asperity contacts occur, causing local oxide growth. Wear debris is from this oxide layer spalling.
 - Regime VII: Metallic contact occurs at asperities (despite the ability of oxide to grow), leading to severe wear through the formation of metallic debris.
 - Regime VIII: Martensite forms at the interface through local heating of asperities followed by quenching through heat conduction into the bulk. This provides local mechanical support of the oxide film because martensite has a high strength, helping to reduce the degree of wear. Wear occurs by the formation of oxide debris.



2. Wear by Hard Particles

Wear by hard particles can be roughly broken down into abrasion and erosion. In three-body abrasive wear material is removed or displaced from a surface by hard particles rolling between two surfaces (b). In two-body abrasion wear is caused by hard protuberances on one of the surfaces (a). In erosion, wear is caused by hard particles striking the surface, either carried by a gas stream or entrained in a flowing liquid. More hard particles may be generated by this process, or by sliding wear, which can result in increasing rates of wear. Abrasion and erosion can be useful in some circumstances, for example grinding and polishing samples for metallographic examination.



Wear Evaluation:

- **Wear Rate:** is volume loss per unit distance and its unit is (m^3/m). It is independent of the load applied. The specific wear rate depends on applied on to cause wear, it is volume loss per unit meter per unit load. Its unit is (m^3/Nm).
- **Coefficient of Friction (μ):** coefficient of friction and is usually symbolized by the Greek letter mu (μ). Mathematically, $\mu = F/N$. where F is the frictional force and N is the normal force.
- **Sliding Wear:**
 - An equation that can be used to deduce the severity of sliding wear, from a simple model. The Archard equation is $Q = KLW/H$. Where Q is the total volume of wear debris produced per unit distance moved, H is the indentation hardness, W is the total normal load, K is a dimensionless constant of proportionality, and L is the sliding length.
 - The higher the applied load / length/ K the higher the wear rate
 - The lower the hardness of the softer material the higher the wear rate
 - From the above equation it is apparent that wear increases linearly with the contact load, K is a measure of the severity of wear and hard materials wear less than soft materials. There is little correlation between K and μ . Furthermore, the simple model does not tell us anything about the mechanism of material removal.

TABLE 32.1 Range of Coefficients of Friction in Metalworking Processes

Process	Coefficient of friction (μ)	
	Cold	Hot
Rolling	0.05–0.1	0.2–0.7
Forging	0.05–0.1	0.1–0.2
Drawing	0.03–0.1	—
Sheet-metal forming	0.05–0.1	0.1–0.2
Machining	0.5–2	—

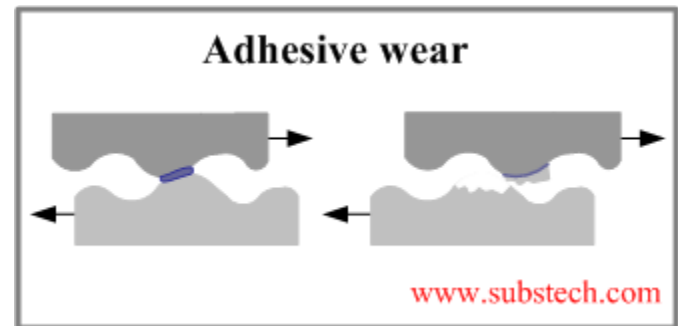
Wear Mechanisms:

It may cause significant surface damage and the damage is usually thought of as gradual deterioration. The following categories are commonly used:

- Adhesive Wear
- Abrasive Wear
- Fretting Wear
- Erosive Wear
- Surface Fatigue
- Cavitation Wear

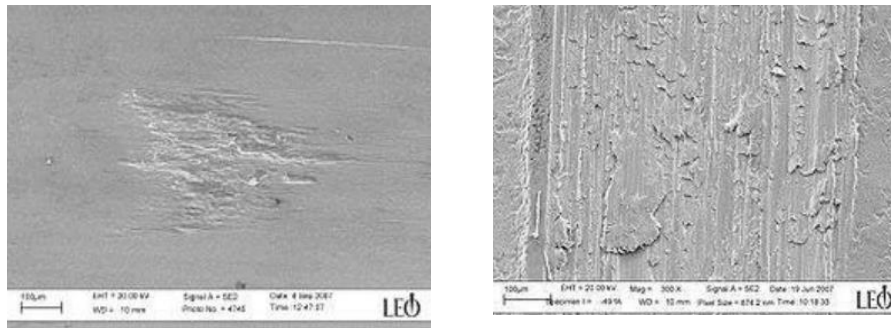
1. Adhesive Wear

- Usually result of removal of an oxide layer
- Brittle oxide layers can create small hard particles that can be entrapped which can cause galling
- Galling can be addition or subtraction
- Lubrication can stop localized welding as it prevents surfaces from being in touch however need to be consistently applied



Adhesive wear may be between metallic materials, ceramics or polymers, or combinations. It is dependent on adhesion between the materials and that in turn depends on surface films like oxides or lubricants, as well as the mutual affinity of one material for another.

- If loads are light and the natural spontaneous oxidation of a metal can keep up with the rate of its removal by wear, then that wear rate will be relatively low (the oxide acting as a lubricant). It is called: Mild Wear. Mild wear is characteristic of dry sliding metals where the conditions are such that the naturally protecting oxide can continuously reform at the sliding contact, so acting with a degree of dry lubrication and reducing the wear rate. It also occurs with hardened alloys (usually steels) when, even under high contact loads and speeds, the underlying substrate can support the oxide and prevent its disruption by deformation below it.
- If loads are high and the protective oxide is continually disrupted to allow intimate metallic contact and adhesion, then the wear rate will be high. It is called: Severe Wear. Severe wear occurs (generally in soft metals or alloys) when the conditions are such that the oxide is disrupted at a greater rate than which it can reform, so that clean metal is exposed below, and massive adhesion occurs between the mating surfaces.
- With materials which have thin, brittle oxides, notably stainless steel, aluminum alloys and titanium, the protective oxide is easily disrupted, and the consequent massive adhesion and wear is called: Galling. Galling is wear caused by adhesion between sliding surfaces. When a material galls, is when parts of the surface are pulled with the contacting surface. Facilitates when a large amount of force compressing the surfaces together.



The terms Mild wear, severe wear and Galling are used with specific meanings. They are in relation to unlubricated sliding and can result in metal surfaces being friction-welded together. Under the right set of circumstances, galling can easily develop between the threads of stainless-steel nuts and bolts, in particular those with coarse threads.

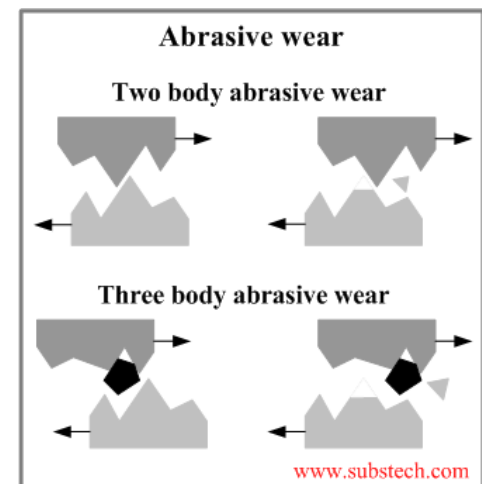
To minimize adhesive wear:

- Select materials that do not form strong adhesive bonds
- Use harder materials as one of the pair
- Use materials that oxidize more easily
- Apply hard coatings that serves functions a and c. Coat one surface with a soft material, such as tin, silver, lead, or cadmium, is also effective in reducing wear. Generally, coatings with high hardness will give lower adhesive wear rates in sliding situations. The hardest coatings are found amongst the thermally sprayed ceramics and cermet, with the best of the electroplates being hard chrome. The selection is application specific, and you should remember to consult the information on Mild and Severe Wear, on Scuffing and on the effect of Lubrication.

2. Abrasive Wear

Abrasive wear occurs when a harder material is rubbing against a softer material.

- If there are only two rubbing parts involved in the friction process the wear is called two body wear.
- In this case the wear of the softer material is caused by the asperities on the harder surface.
- If the wear is caused by a hard particle (grit) trapped between the rubbing surfaces it is called three body wear. The particle may be either free or partially embedded into one of the mating materials.



In the micro-level abrasive action results in one of the following wear modes:

- Ploughing.** The material is shifted to the sides of the wear groove. The material is not removed from the surface.
- Cutting.** A chip forms in front of the cutting asperity/grit. The material is removed (lost) from the surface in the volume equal to the volume of the wear track (groove).

- Cracking (brittle fracture). The material cracks in the subsurface regions surrounding the wear groove. The volume of the lost material is higher than the volume of the wear track.

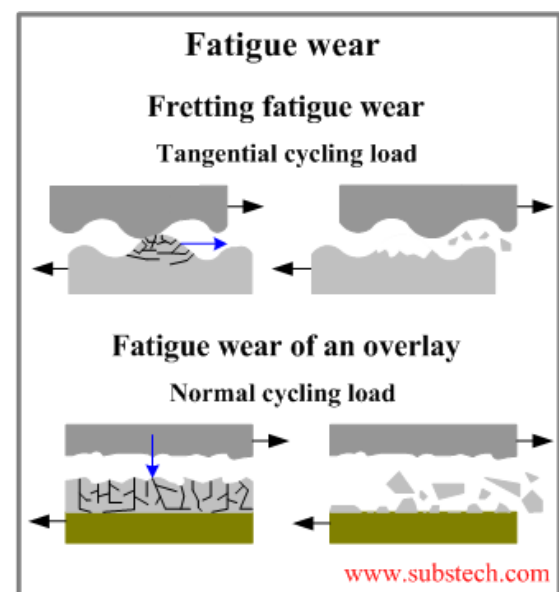
Factors Influencing Abrasive Wear:

Several different mechanisms have been proposed to describe the manner in which the material is removed. Three commonly identified mechanisms of abrasive wear are:

- **Plowing:** Plowing/Ploughing occurs when material is displaced to the side, away from the wear particles, resulting in the formation of grooves that do not involve direct material removal. The displaced material forms ridge adjacent to grooves, which may be removed by subsequent passage of abrasive particles.
- **Cutting:** Cutting occurs when material is separated from the surface in the form of primary debris, or microchips, with little or no material displaced to the sides of the grooves. This mechanism closely resembles conventional machining.
- **Fragmentation:** Fragmentation occurs when material is separated from a surface by a cutting process and the indenting abrasive causes localized fracture of the wear material. These cracks then freely propagate locally around the wear groove, resulting in additional material removal by spalling.

3. Fatigue Wear

Surface fatigue Wear: Also called Surface fatigue. Surface fatigue is a process by which the surface of a material is weakened by cyclic loading, which is one type of general material fatigue. Fatigue wear is produced when the wear particles are detached by cyclic crack growth of microcracks on the surface. These microcracks are either superficial cracks or subsurface cracks. Repeated, alternating mechanical stresses lead to the formation and propagation of cracks under the stressed surface, which is thus destroyed. The consequences are transverse and crest cracks, pitting and micro-pitting (especially in rolling contact), tool breakage.

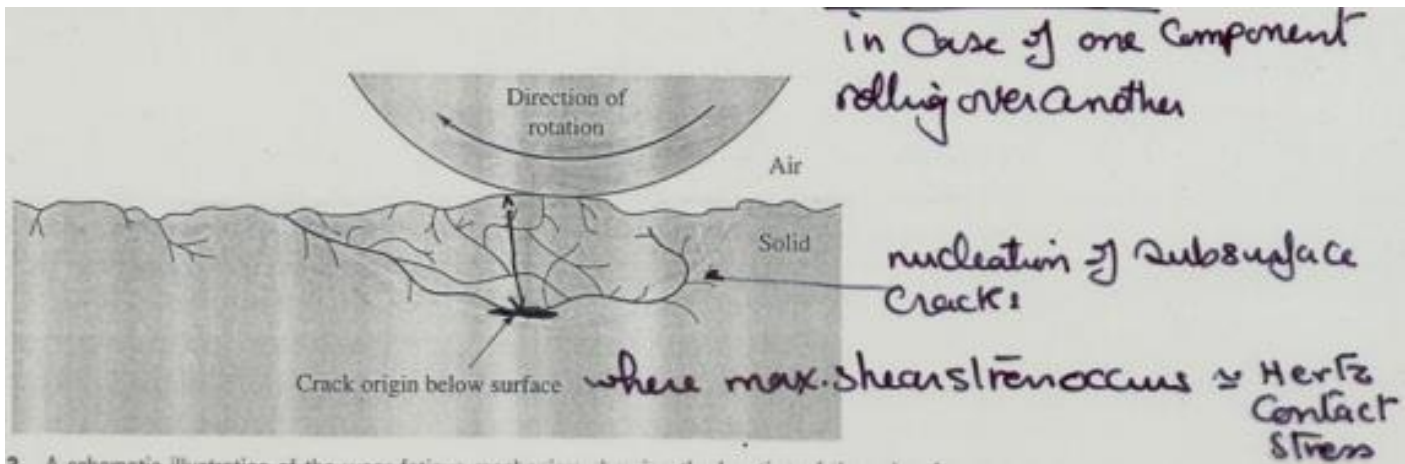


- Fatigue wear of a material is caused by a cycling loading during friction. Fatigue occurs if the applied load is higher than the fatigue strength of the material.
- Fatigue cracks start at the material surface and spread to the subsurface regions. The cracks may connect to each other resulting in separation and delamination of the material pieces.
- One of the types of fatigue wear is **fretting wear** caused by cycling sliding of two surfaces across each other with a small amplitude (oscillating). The friction force produces alternating compression-tension stresses, which result in surface fatigue.
- Fatigue of overlay of an engine bearing may result in the propagation of the cracks up to the intermediate layer and total removal of the overlay.

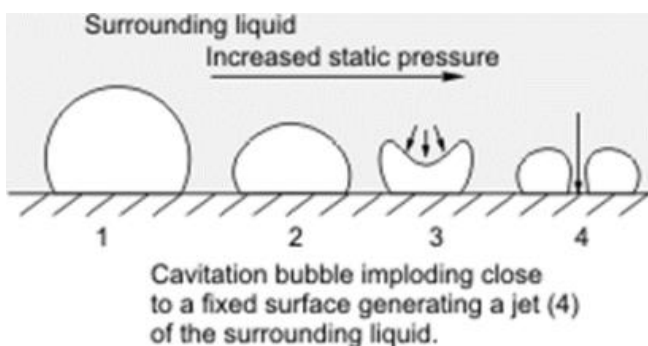
Fatigue Wear during Rolling:

During rolling, the local contact stresses are very high, which are concentrated over a small area and are repetitive. Such loading occurs in rolling bearings, gears, friction drives, cams, and followers, etc. Steps leading to generation of wear particles are:

- Application of normal load that induces stresses at contact points.
- Growth of plastic deformation per cycle.
- Subsurface crack nucleation.
- Expansion of crack due to reversal of stress.
- Extension of crack to the surface due to traction force.
- Generation of wear particles.



4. Cavitation Wear

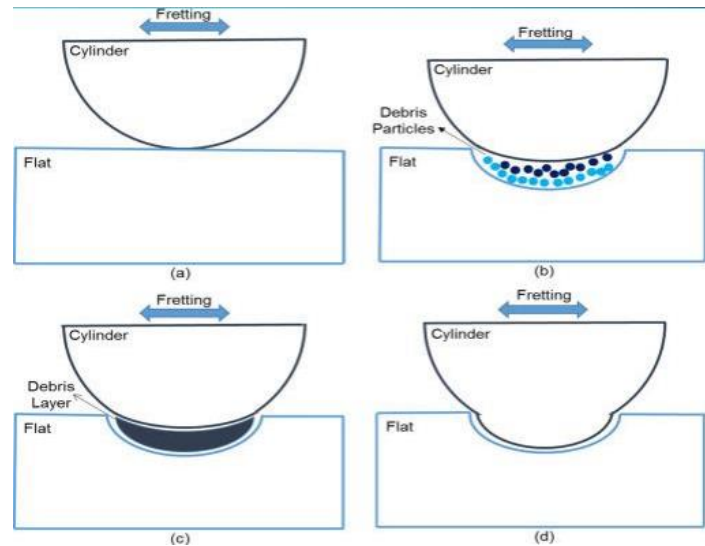


- Caused by fluid flowing through a system
- If there is a point with high turbulence results in creating of bubbles which hits the surface of the pipe and explodes, and when this happens heat is generated, and surface is eroded.
- Burned/Darkened surface is a characteristic of cavitation wear
- Very rough surface.

5. Fretting Wear

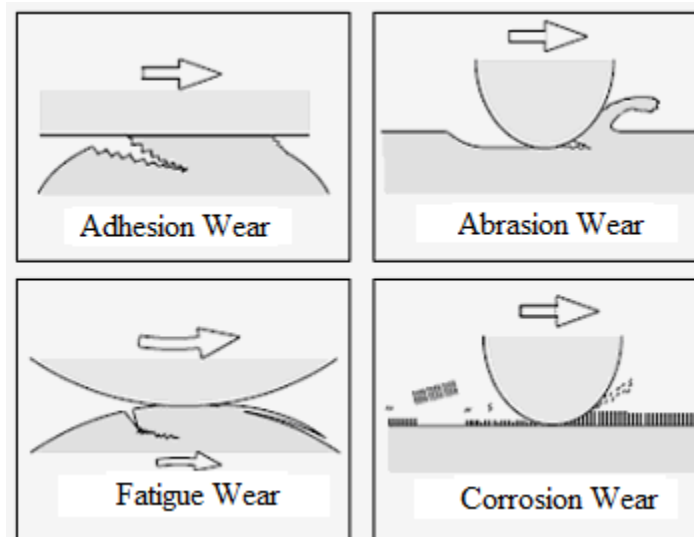
Fretting wear is the repeated cyclical rubbing between two surfaces, which is known as fretting, over a period of time which will remove material from one or both surfaces in contact. It occurs typically in bearings, although most bearings have their surfaces hardened to resist the problem.

- Asperities later get welded to one region due to heat generated
- Caused by fragmentation problem



6. Corrosive Wear

When rubbing surfaces removes an oxide layer of one of the sliding surfaces. Wear in a corrosive environment which increases wear rate.

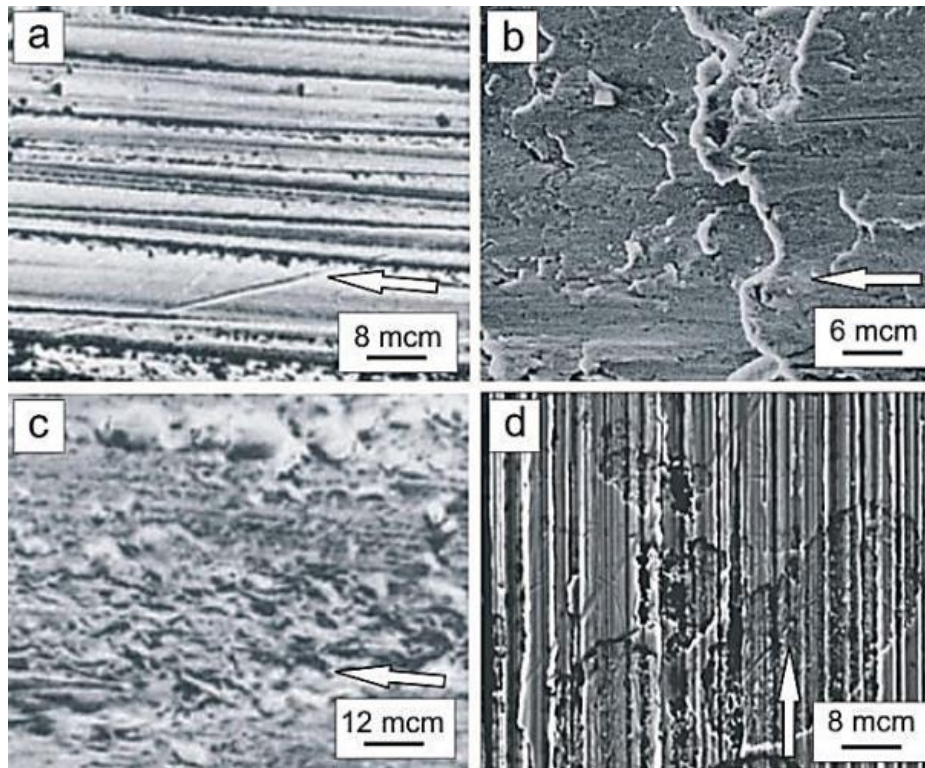


Characterization and Modeling of the Wear Situation

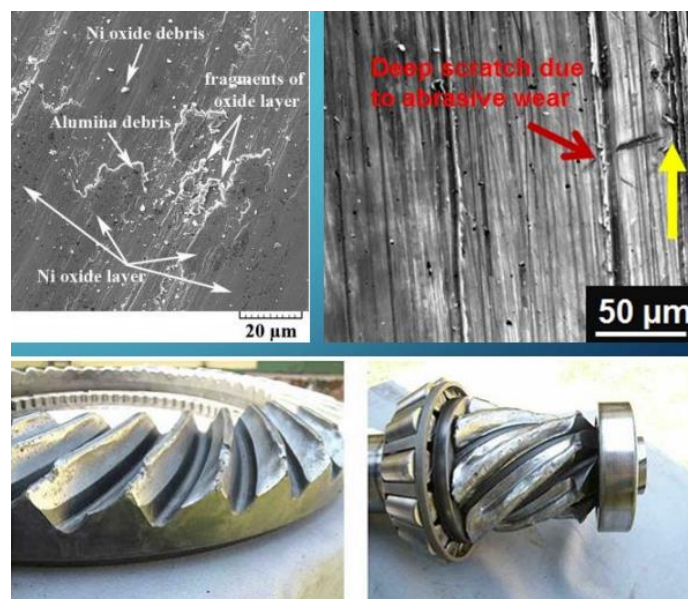
The term abrasive wear is used to describe situations where the principal cause of the wear is scratching or cutting by abrasive particles. The term nonabrasive wear is used to describe all other wear situations involving contact between two solid bodies (for example, sliding). The term erosion is used where the wear is caused by a fluid, a stream of particles, or bubbles (in the case of cavitation), not by contact between two solid bodies. The operational classification for nonabrasive wear situations (Table 1) is directly based on the characterization of the elements of the tribosystem at the contact.

Table 1 Operational attributes of nonabrasive wear

Attribute	Variations
Motion	
Rolling	With slip Without slip
Impact	With slip Without slip
Sliding	Unidirectional or reciprocating High speed or low speed Fretting or gross sliding
Lubrication	Lubricated or not Type of lubricant Lubricant
Load	Heavy or light Constant or variable
	Impact or not
Contact geometry	Point, line, or area Conforming or nonconforming
Contact stress	Above or below yield
Environment	Hostile or nonhostile High temperature or low temperature With or without abrasive particles pH level
Materials	Type-to-type or dissimilar

EXAMPLES:

- A is ploughing (type of abrasive)
- B is surface being plastically deformed due to wear which can be due to material softening due to head
- C shows ploughing and adhesive wear
- D shows fatigue wear which can be seen by removed section and indentations



- Top right image shows a large particle that was entrapped in between
- Both gears need higher magnification to know which type